

how a fluent black box becomes a verified design

same scoring, every system:

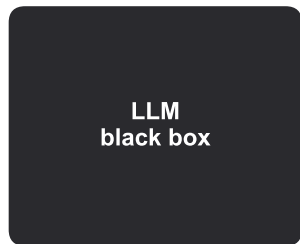
bare LLM 0–12% usable · hallucination ~0.9–1.0

Labwright 88–100% usable · hallucination 0.000

fluency without derivation

the gate — calculators + verifier

determinism — recomputed & verified



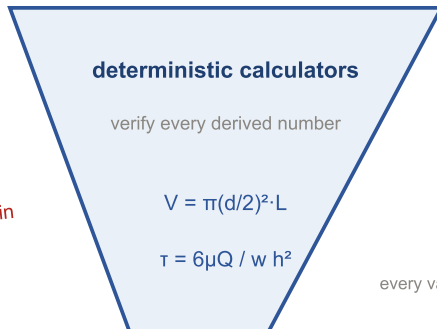
raw inputs only

$$\tau = 0.05 \text{ Pa}$$

$$V = 2.9 \text{ }\mu\text{L}$$

$$d = 100 \text{ }\mu\text{m}$$

$$Q = 2 \text{ }\mu\text{L/min}$$

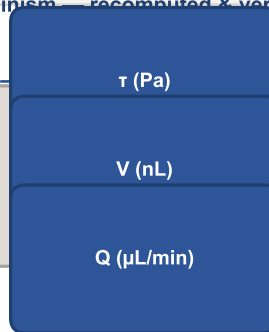
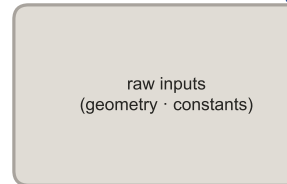


$$P(\text{Solver Error}) \equiv 0$$



broken
chemistry

no derivation, no entry —
the gate rejects typed numbers



every value follows from raw inputs alone



honest boundary: the gate verifies
arithmetic, not physiology — on blind
goals usable drops, hallucination 0.000